ACTIVITY - 1

1. Medical Image Enhancement

Scenario: A x-ray image shows bones clearly, but darker tissues have poor visibility.

1. Which grey-level transformation would you recommend?

I would recommend contrast stretching to improve the contrast of the x-ray image. It helps utilize a wider range of intensity values.

2. Why it is suitable?

Contrast stretching expands the range of intensity values, making differences between dark and bright regions more visible. It is useful when the original image has a limited range of grey levels.

3. What effect it will have on the image?

The image will have improved contrast, and the tissues will become easier to distinguish from surrounding regions. However, the result depends on the original intensity distribution.

2. Night-time Surveillance

Scenario: Most pixel values in a CCTV image are between 0 and 40, making the image very dark

1. Which transformation would improve visibility.

I would recommend Logarithmic Transformation (Log Transformation).

2. Why is it more suitable than negative transformation

Log transformation expands low-intensity pixel values, making dark regions more visible. Negative transformation reverses brightness levels but does not specifically enhance dark intensity differences in the same way.

3. Predict the appearance of enhanced image.

The parking lot will appear brighter, and objects such as parked vehicles, people and surrounding structures may become more visible.

3. Finger print Recognition

Scenario: A finger print image contains ridge details, but the contrast between ridges and valleys is low.

1. Which grey transformation would you apply.

I would apply contrast streching.

2. Explain why?

Contrast stretching expands the intensity range and increases the difference between ridge and valley pixels, making fingerprint patterns easier to distinguish.

3. What would happen if a log transformation were applied instead

Log transformation would enhance difference among darker pixel values, potentially improving some ridge details. However, it may not improve the overall ridge-valley contrast as effectively as contrast streching if the image's intensity range is poorly utilized.